

Roll No. 220010286023

61475

**B. Pharma. 6th Semester
(Regular/Re-appear/Improvement)
Examination – May-2025**

MEDICINAL CHEMISTRY III

Paper : BP601T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt *all* questions from Section-A & any *two* questions from Section-B and any *Seven* questions from Section-C.

SECTION – A

1. Attempt *all* questions : 2 × 10 = 20
- (a) What are Cephalosporins.
 - (b) Draw the structure of Amantadine and give its use.
 - (c) Name the agents used in the treatment of TB.

P. T. O.

61475

- (d) Draw structure of Chloroquin and give its use.
- (e) Introduce Penicillins with *two* examples.
- (f) Write structure and use of Sulphacetamide.
- (g) Define Prodrugs ?
- (h) What are antimalarial agents ?
- (i) Define Beta Lactam Antibiotics.
- (j) What are Antiprotozoal drugs ? Give any two examples.

SECTION – B

2. Long Answer Questions (Attempt any *two*) : $10 \times 2 = 20$

- (a) Explain in detail about QSAR.
- (b) Discuss the SAR quinolines, MOA and uses.
- (c) Explain Application of Prodrugs.

SECTION – C

3. Short Answer Questions (Attempt any *Seven*) :

$$7 \times 5 = 35$$

- (a) Discuss the SAR of Tetracyclines with suitable examples.
- (b) Classify anti-tubercular agents.

- (c) Discuss the chemical structure MOA and synthesis of Isoniazid.
 - (d) Describe Anthelmintics giving examples.
 - (e) Discuss in detail about anti- TB agents.
 - (f) Write a note on Docking techniques.
 - (g) Write the mechanism of action and synthesis of Ciprofloxacin.
 - (h) Write a note on Combinatorial Chemistry.
 - (i) Write the mechanism of action and synthesis of Sulphamethoxazole.
-

DEPARTMENT OF PHARMACEUTICAL SCIENCES, INDIRA GANDHI UNIVERSITY, MEERPUR, REWARI

First Sessional Theory Examination of B. Pharmacy 6th Semester –February, 2025

Subject: Medicinal Chemistry - III Theory (Paper Code: BP601T)

(Exam Date: 27.02.2025)

Time: 1 hr

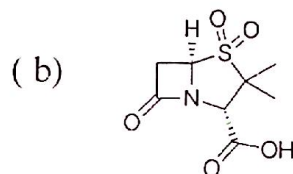
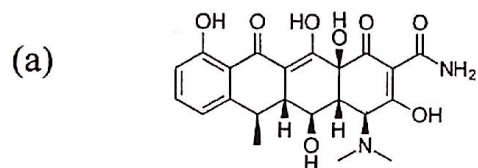
Max. Marks =30

Attempt all questions

Q1. Give the answers of all questions -

5 × 2 =10

- i. Define antibiotic with example
- ii. Sketch the molecular structure of β -Lactam ring with IUPAC name.
- iii. Identify the name of given structures
- vi. Write down the two examples of Macrolide antibiotics
- v. Write the synthesis of Chloramphenicol.



Q2. Attempt any two question (Short answer type questions)

2 × 5 = 10

- a) Discuss the history of antibiotic with suitable examples.
- b) Discuss the structure activity relationship of quinolones.
- c) What is the degradation? Discuss the degradation of penicillin.

Q3. Attempt any one question (Long answer type questions)

1 × 10 = 10

- a) What is the tetracycline? Discuss the mechanism and nomenclature of tetracycline with suitable drug example.
- b) What are the prodrugs? Discuss the type and applications of prodrug design.

Roll No _____

DEPARTMENT OF PHARMACEUTICAL SCIENCES, INDIRA GANDHI UNIVERSITY, MEERPUR, REWARI

Second Sessional Theory Examination of B. Pharmacy 6th Semester –April, 2025

Subject: Medicinal Chemistry - III Theory (Paper Code: BP601T)

(Exam Date: 21.04.2025)

Max. Marks =30

Time: 1 hr

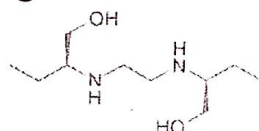
Attempt all questions

Q1. Give the answers of all questions -

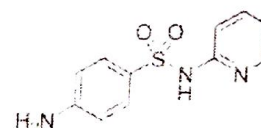
5 × 2 =10

- | | |
|---|--|
| i. Write the uses of Sparfloxacin and Tolnaftate drugs. | iv Write down the mechanism action of Antifungal agents. |
| ii. Sketch the molecular structure of Cotrimoxazole. | v Write the synthesis of Nitrofurantoin drug. |
| iii. Write the name of given structures | |

(a)



(b)



Q2. Attempt any two question (Short answer type questions)

2 × 5 = 10

- Discuss the classification of antiviral agents with molecular structure.
- Discuss the structure activity relationship of Sulphonamides with suitable examples.
- Write the synthesis of Isoniazid and Dapsone drugs.

Q3. Attempt any one question (Long answer type questions)

1 × 10 = 10

- Discuss the drug profile (including about the molecular structure, mechanism of action, and uses) of Pyrazinamide, Ofloxacin, Zidovudine and Thiabendazole drugs.
- Write the synthesis of Mebendazole and Ciprofloxacin drugs in details